PLAN 372 HW 6

Street trees provide a host of benefits to urban environments. They create shade, improve air quality, slow traffic on neighborhood streets, and much more. The City of Raleigh wishes to update its development standards to make recommendations and guide investment in street trees. One of the key questions they have is about what types of trees work well to maximize shade, and has engaged you as a consultant to study this question.

They have recommended you use data from US Department of Agriculture's Urban Tree Database. This dataset contains data on over 14,000 urban trees in the US. Download the "data publication" from that link; within it you will find several data files. For this study, use the "Raw Tree Data," which contains individual records for each tree studied.

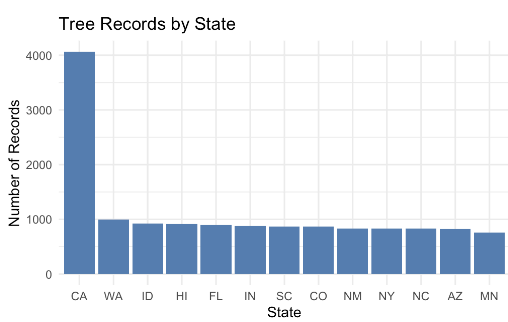
You can accept the assignment on Github Classroom through this link: https://classroom.github.com/a/twM98TBo. This homework does *not* need to be in report format; you can just provide an answer to each question.

Question 1: Sample sizes by state

The dataset does not contain a state column, but does contain a city column which contains both city and state information (e.g. Charlotte, NC). Use a regular expression to create separate columns for the city name and state abbreviation, and use these columns to answer the following questions.

How many records are there in each state (include a table or bar plot)? [3 points]

```
> print(state_co
   state    n
1     CA 4062
2     WA  994
3     ID  923
4     HI  918
5     FL  895
6     IN  877
7     SC  872
8     CO  867
9     NM  833
10    NY  831
11    NC  828
12    AZ  827
13    MN  760
>
```



Question 2: Cities in NC/SC

Since different trees grow differently in different parts of the country, the city wants you to only use data from North and South Carolina. Filter the dataset to only these states, and **use that filtered dataset for all remaining questions.**

What cities did they collect data from in North and South Carolina? [2 points]

Charlotte and Charleston

Question 3: Genera and species

The city wishes to know what types of trees in the dataset have the largest crown (i.e. their branches and leaves cover the most area, maximizing shade). The crown size is in the column `AvgCdia (m)`, in meters.

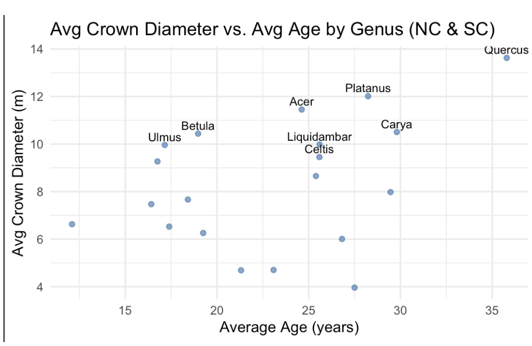
The data contain the scientific names of the species of each surveyed tree. Scientific names use binomial nomenclature, where the scientific names contain both a genus (plural genera) and species. For instance, *Acer saccharum* is the sugar maple; the genus is *Acer* and the species is *saccharum*. Trees within a genus are related and may share physical characteristics. For instance, *Acer rubrum* or the red maple is another large maple tree. To maximize sample size, the city has requested you compute the average canopy diameter for each genus. This requires you to write a regular expression to extract the genus.

What genus of trees has the largest average crown diameter in North and South Carolina, and what is the diameter? [3 points]

Quercas

Question 4: Tree age

Older trees, of course, have larger crowns. Are there differences in the average age of the different genera of trees in the dataset? Might this explain the results of the previous question? [2 points]



Yes, there are differences in average age among the genera. Quercus is the oldest at about 35 years and has the largest crown. In contrast, genera like Ulmus and Betula are much younger, around 17 to 19 years old, but still have fairly large crowns. However, age alone does not fully explain the pattern, because some younger genera like Ulmus and Betula also have large crowns.

AI reflection

Include a few sentences at the end of your assignment discussing any AI tools or outside resources you used (e.g. ChatGPT, StackOverflow, Google AI), how you used them, and whether you felt they supported your learning or not [0 points, but I will send the assignment back for you to add this if you don't include it].

Extra credit

Genera recommendation

Recommend a genera that produces a large crown quickly (among trees in North and South Carolina). You can use any analytical methods you want (group by, plots, linear regression, etc.). Document the process by which you chose this type of tree. [1 point]

Ulmus is the best choice because it has the highest crown-per-year ratio at 0.58 meters per year. This means its canopy grows faster than any other genus, and it still reaches a large final crown size of about 10 meters.

Species

So far, all of the analysis has focused on genera. Refine your regular expression to also extract the species as well, as a separate column. Within each genus of tree in the entire dataset (not just North and South Carolina), how many species are recorded in the dataset? [1 points] **143**

Most of the time, scientific names are just *Genus species*, but some are more complicated. Be sure to account for the following cases, and extract just the species:

- Hybrid plants may have an x in between the genus and species.
  - For example, for "Platanus x acerifolia" or the London Plane, the genus name is "Platanus" and the species name is "acerifolia"; your regular expression should remove the x
- Some plants may have additional information after the species.
  - For instance, "Carpinus betulus 'Fastigiata'", the genus is "Carpinus" and the species is "betulus"—Fastigiata is a *cultivar* which your regular expression should remove.

- “Juniperus virginiana var. silicicola”, the genus is “Juniperus” and the species is “virginiana”—Silicolica is a *variety* which your regular expression should remove.